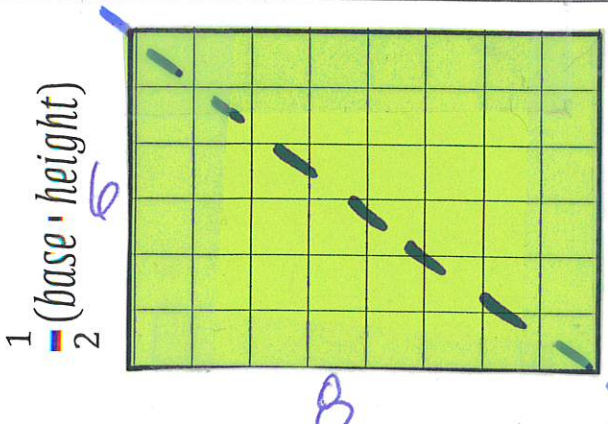
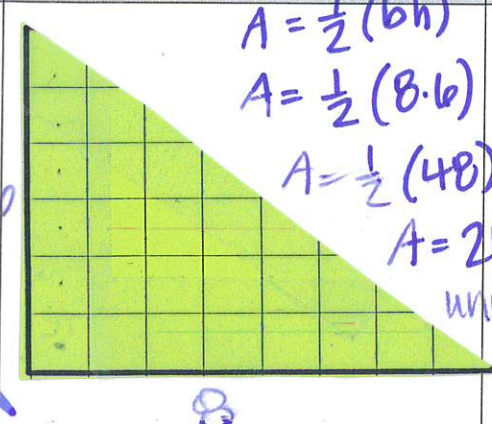
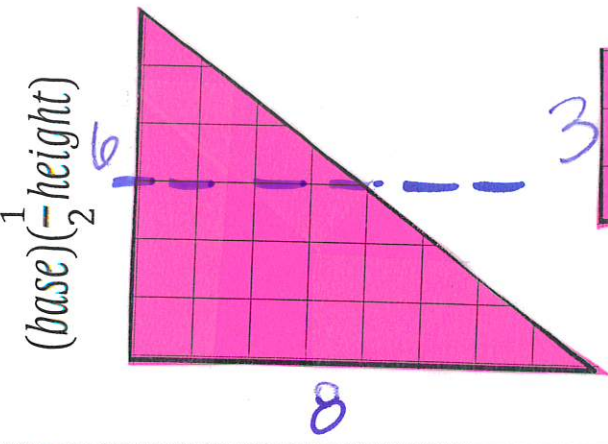
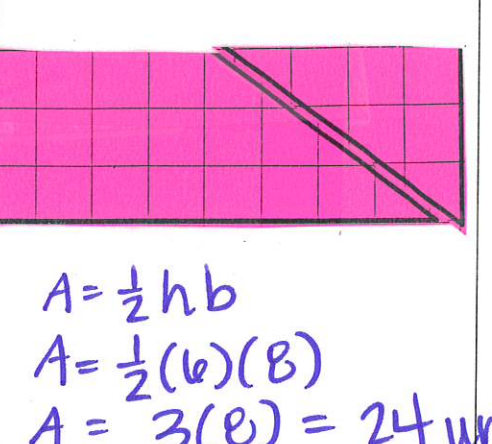


Unit 9: Geometric Equations Day 2: Decomposing and Rearranging Triangles

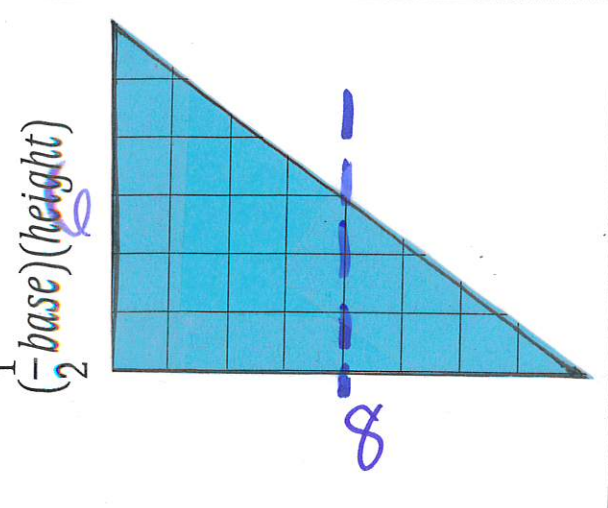
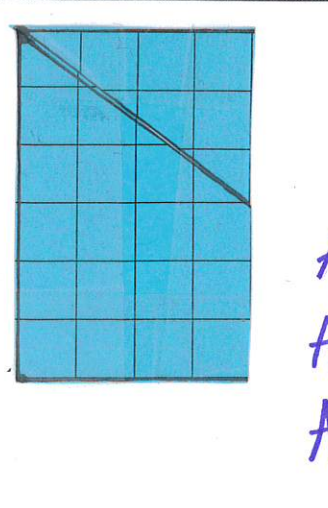
Cut the area of the RECTANGLE in HALF on the diagonal.

Original shape	Decomposed shape	Describe how to find the area
 $\frac{1}{2}(\text{base} \cdot \text{height})$ 6 8	 $A = \frac{1}{2}(bh)$ $A = \frac{1}{2}(8 \cdot 6)$ $A = \frac{1}{2}(48)$ $A = 24$ units²	Multiply the <u>base</u> and <u>height</u> of the rectangle. Then multiply by <u>half</u>. This is the same as dividing by <u>2</u>.

Cut the HEIGHT of the triangle in HALF.

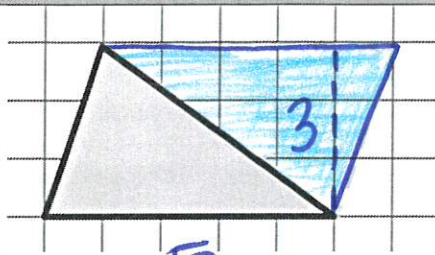
Original shape	Decomposed shape	Describe how to find the area
 $(\text{base})(\frac{1}{2}\text{height})$ 6 8	 3 $A = \frac{1}{2}hb$ $A = \frac{1}{2}(6)(8)$ $A = 3(8) = 24$ units²	Multiply <u>half</u> of the <u>height</u> by the <u>base</u>.

Cut the BASE of the triangle in HALF.

Original shape	Decomposed shape	Describe how to find the area
 $(\frac{1}{2}\text{base})(\text{height})$ 6 8	 $A = \frac{1}{2}bh$ $A = \frac{1}{2}(8)(6)$ $A = 4(6) = 24$ units²	Multiply <u>half</u> of the <u>base</u> by the <u>height</u>.

Draw a second triangle to create a rectangle or parallelogram.
Find the area of the triangle.

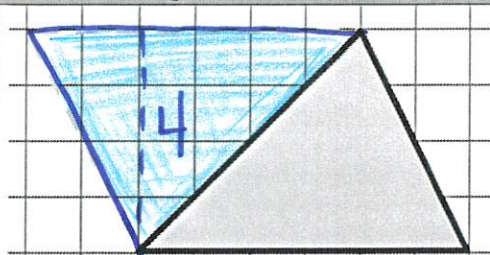
$$\frac{1}{2}(\text{base} \cdot \text{height})$$



$$A = \frac{1}{2}bh$$

$$A = \frac{1}{2}(5)(3)$$

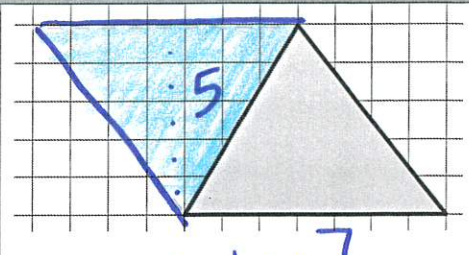
$$A = \frac{1}{2}(15) = 7.5 \text{ units}^2$$



$$A = \frac{1}{2}bh$$

$$A = \frac{1}{2}(6)(4)$$

$$A = \frac{1}{2}(24) = 12 \text{ units}^2$$



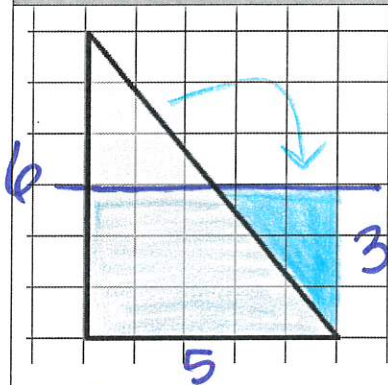
$$A = \frac{1}{2}bh$$

$$A = \frac{1}{2}(7)(5)$$

$$A = \frac{1}{2}(35) = 17.5 \text{ units}^2$$

Cut the HEIGHT of the triangle in HALF. Draw the rearranged rectangle.

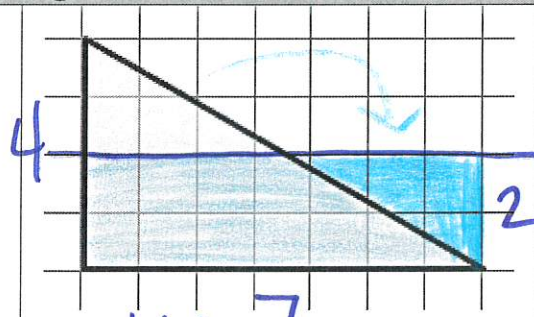
$$(\text{base})\left(\frac{1}{2}\text{height}\right)$$



$$A = \frac{1}{2}hb$$

$$A = \frac{1}{2}(6)(5)$$

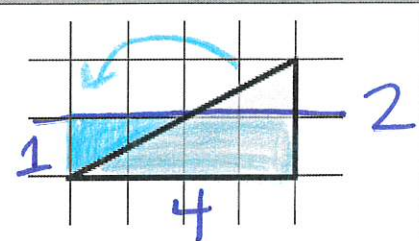
$$A = 3(5) = 15 \text{ units}^2$$



$$A = \frac{1}{2}hb$$

$$A = \frac{1}{2}(4)(7)$$

$$A = 2(7) = 14 \text{ units}^2$$



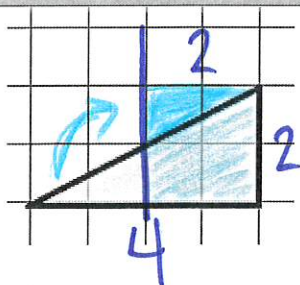
$$A = \frac{1}{2}hb$$

$$A = \frac{1}{2}(2)(4)$$

$$A = 1(4) = 4 \text{ units}^2$$

Cut the BASE of the triangle in HALF. Draw the rearranged rectangle.

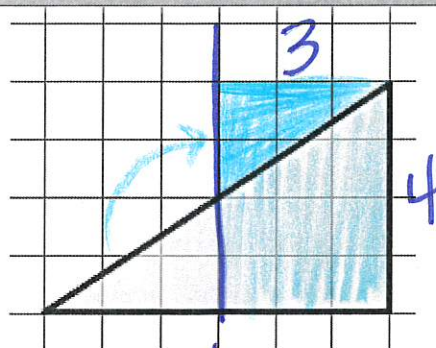
$$\left(\frac{1}{2}\text{base}\right)(\text{height})$$



$$A = \frac{1}{2}bh$$

$$A = \frac{1}{2}(4)(2)$$

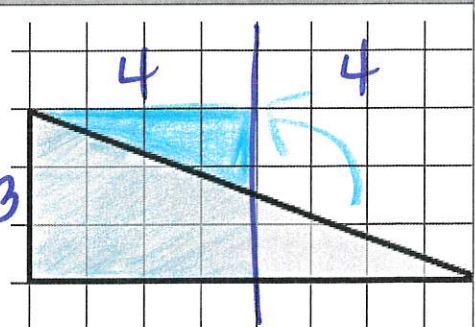
$$A = 2(2) = 4 \text{ units}^2$$



$$A = \frac{1}{2}bh$$

$$A = \frac{1}{2}(6)(3)$$

$$A = 3(3) = 9 \text{ units}^2$$



$$A = \frac{1}{2}bh$$

$$A = \frac{1}{2}(8)(3)$$

$$A = 4(3) = 12 \text{ units}^2$$